

DATABASE DESIGN DOCUMENT

AI-Based Smart Learning Assistant

Implemented Schema — Week 5 Backend Development

CSE4104-7A-T01
Engine: SQLite (dev)

v1.0
Jun 2026

11 Tables
FK Enforced

1. OVERVIEW

This document describes the database schema implemented for the AI-Based Smart Learning Assistant backend, matching the ER Diagram approved in Week 4. All 11 entities, their fields, primary keys, foreign keys, and check constraints are detailed below.

- Engine (local dev/grading): SQLite via Node.js built-in node:sqlite module
- Engine (production target): PostgreSQL — schema is provider-agnostic standard SQL
- Foreign key enforcement: PRAGMA foreign_keys = ON
- Primary keys: UUID strings (generated via the uuid package) for all tables
- Enum-like fields enforced via SQL CHECK constraints (e.g., status, priority, role)
- JSON-array fields (e.g., tags) stored as TEXT and serialized/deserialized at the application layer

2. ENTITY DEFINITIONS

users

Student and admin accounts — authentication, profile, study goal tracking

Column	Type	Constraints / Notes
id	TEXT	PRIMARY KEY (UUID)
email	TEXT	UNIQUE NOT NULL
name	TEXT	NOT NULL
password	TEXT	bcrypt hash, salt factor 12; nullable for OAuth-only users
google_id	TEXT	UNIQUE — reserved for Google OAuth integration
role	TEXT	CHECK IN ('STUDENT','ADMIN'), DEFAULT 'STUDENT'
streak	INTEGER	DEFAULT 0 — consecutive active study days
study_goal_hrs	REAL	DEFAULT 4
email_verified	INTEGER	0/1 boolean flag
created_at / updated_at	TEXT	ISO 8601 timestamps

refresh_tokens

Persisted JWT refresh tokens with expiry, enabling token revocation

Column	Type	Constraints / Notes
id	TEXT	PRIMARY KEY (UUID)
token	TEXT	UNIQUE NOT NULL
user_id	TEXT	FOREIGN KEY → users.id, ON DELETE CASCADE
expires_at	TEXT	ISO 8601 — checked on every refresh request

subjects

Catalog of academic subjects (Math, Physics, CS, etc.)

Column	Type	Constraints / Notes
id	TEXT	PRIMARY KEY (UUID)
name	TEXT	UNIQUE NOT NULL
color / icon	TEXT	UI display attributes

user_subjects

Many-to-many: tracks each user's proficiency per enrolled subject

Column	Type	Constraints / Notes
id	TEXT	PRIMARY KEY (UUID)
user_id	TEXT	FOREIGN KEY → users.id, ON DELETE CASCADE
subject_id	TEXT	FOREIGN KEY → subjects.id
proficiency	INTEGER	0-100 scale, DEFAULT 50
(user_id, subject_id)	—	UNIQUE composite constraint

tasks

Core task management — status, priority, due dates, time tracking

Column	Type	Constraints / Notes
id	TEXT	PRIMARY KEY (UUID)
title	TEXT	NOT NULL
status	TEXT	CHECK IN ('TODO','IN_PROGRESS','DONE','CANCELLED')
priority	TEXT	CHECK IN ('LOW','MEDIUM','HIGH','URGENT')
due_date	TEXT	Nullable ISO 8601 timestamp
tags	TEXT	JSON-encoded array, DEFAULT '[]'
user_id	TEXT	FOREIGN KEY → users.id, ON DELETE CASCADE
subject_id	TEXT	FOREIGN KEY → subjects.id, nullable
completed_at	TEXT	Set when status transitions to DONE

study_plans

AI-generated or manually created study plans

Column	Type	Constraints / Notes
id	TEXT	PRIMARY KEY (UUID)
status	TEXT	CHECK IN ('ACTIVE','COMPLETED','ARCHIVED')
start_date / end_date	TEXT	ISO 8601 — plan duration window
user_id	TEXT	FOREIGN KEY → users.id, ON DELETE CASCADE

chat_sessions		
AI chatbot conversation threads, one per topic		
Column	Type	Constraints / Notes
id	TEXT	PRIMARY KEY (UUID)
title	TEXT	DEFAULT 'New Chat' — auto-set from first message
user_id	TEXT	FOREIGN KEY → users.id, ON DELETE CASCADE

chat_messages		
Individual messages within a chat session (user + AI assistant)		
Column	Type	Constraints / Notes
id	TEXT	PRIMARY KEY (UUID)
session_id	TEXT	FOREIGN KEY → chat_sessions.id, ON DELETE CASCADE
role	TEXT	CHECK IN ('USER','ASSISTANT','SYSTEM')
content	TEXT	NOT NULL — message body
tokens	INTEGER	Nullable — AI response token count for usage tracking

recommendations		
AI-curated learning resources (videos, articles, courses, etc.)		
Column	Type	Constraints / Notes
id	TEXT	PRIMARY KEY (UUID)
type	TEXT	CHECK IN ('VIDEO','ARTICLE','EXERCISE','BOOK','PODCAST','COURSE')
saved	INTEGER	0/1 — user bookmark flag
user_id / subject_id	TEXT	FOREIGN KEYS → users.id / subjects.id

study_logs		
Logged study sessions used for dashboard analytics and streaks		
Column	Type	Constraints / Notes
id	TEXT	PRIMARY KEY (UUID)
duration_min	INTEGER	NOT NULL
user_id / subject_id	TEXT	FOREIGN KEYS → users.id / subjects.id
date	TEXT	ISO 8601 — used for weekly aggregation queries

notifications		
In-app notifications (deadline reminders, system alerts)		
Column	Type	Constraints / Notes
id	TEXT	PRIMARY KEY (UUID)
read	INTEGER	0/1 boolean flag, DEFAULT 0
user_id	TEXT	FOREIGN KEY → users.id, ON DELETE CASCADE

3. RELATIONSHIP SUMMARY

- users (1) $\leftrightarrow$ (N) tasks, study_plans, chat_sessions, recommendations, study_logs, notifications, refresh_tokens
- users (M) $\leftrightarrow$ (N) subjects, via user_subjects junction table
- subjects (1) $\leftrightarrow$ (N) tasks, recommendations, study_logs
- chat_sessions (1) $\leftrightarrow$ (N) chat_messages

This relationship structure is identical to the ER Diagram submitted in Week 4 — no entities or relationships were added, removed, or restructured during implementation.